Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

Sea Surface Temperature Production Centre

Mediterranean Sea and Black Sea

Sea Surface Temperature Reprocessing

SST_MED_SST_L4_REP_OBSERVATIONS_010_021
SST_BS_SST_L4_REP_OBSERVATIONS_010_022
SST_BS_PHY_L3S_MY_010_041
SST_MED_PHY_L3S_MY_010_042

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Approval date by the CMEMS product quality coordination team: 15/11/2024

CHANGE RECORD

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	10/2/2015	All	Reformatted for MYOFO	B. Buongiorno Nardelli	
1.1	05/03/2015	2, 3, 5	Modified after QUARG review	B. Buongiorno Nardelli, A. Pisano	
1.2	01/09/2015	All	Change format to fit CMEMS graphical rules. Introduce references to CMEMS.	B. Hackett	L. Crosnier
1.3	26/08/2016	All	Added description of interim REP data for recent years	B. Buongiorno Nardelli, A.Pisano	
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1.5	03/04/2020	All	Update to ESA CCI SST/C3S v.2.0	A. Pisano, B. Buongiorno Nardelli, C. Cesarini	Mercator Ocean
1.6	14/12/2021	All	Automatic temporal extension	A. Pisano, C. Fanelli, C. Cesarini, B. Buongiorno Nardelli	A. Pisano
1.7	11/04/2022	II.1.1, II.1.5	Upstream data change	A. Pisano, C. Fanelli, C.Tronconi, C.Cesarini, B.Buongiorno Nardelli	A. Pisano
2.0	16/06/2022	All	Release of MED-L3S-REP and BS-L3S-REP products	A. Pisano, C. Fanelli, C.Tronconi, C.Cesarini, Flavio La Padula, B.Buongiorno Nardelli	A. Pisano
3.0	16/06/2023	I.1, II.1.1, II.1.3, II.1.5, V	Automatic temporal extension	A. Pisano, C. Fanelli, C. Tronconi, C. Cesarini, Flavio La Padula, B.Buongiorno Nardelli	A. Pisano
4.0	31/05/2024	All	Full reprocessing based on ESA CCI SST v.3.0	A. Pisano, C. Fanelli, A. Massi, C. Tronconi, C. Cesarini, F. La Padula, B. Buongiorno Nardelli, D. Ciani	A. Pisano

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I EXECUTIVE SUMMARY

I.1 Products covered by this document

This document describes the quality of the Copernicus Marine Service Mediterranean Sea and Black Sea reprocessed (REP) sea surface temperature (SST) products (Table 1), developed for climate applications by the Italian National Research Council - Institute of Marine Sciences (CNR-ISMAR).

Table 1. *The Copernicus Marine Service Mediterranean Sea (MED) and Black Sea (BS) reprocessed (REP) merged multi-sensor (L3S) and optimally interpolated (L4) SST data. These products provide daily foundation SST maps for each basin, spanning from 1982 to the present, with a spatial resolution of approximately 5 km.*

Short Description	Product Reference	Area	Delivery Time	DOI
MED REP L4	SST_MED_SST_L4_REP_OBSERVATIONS_010_021	MED	Daily[1]	10.48670/moi-00173
MED REP L3S	SST_MED_PHY_L3S_MY_010_042	MED	Daily	10.48670/moi-00314
BS REP L4	SST_BS_SST_L4_REP_OBSERVATIONS_010_022	BS	Daily	10.48670/moi-00160
BS REP L3S	SST_BS_PHY_L3S_MY_010_041	BS	Daily	10.48670/moi-00313

These products are a collection of daily (night time) merged multi-sensor Level-3S (L3S) and optimally interpolated Level-4 (L4) satellite-based estimates of the foundation SST (i.e. the temperature free - or nearly-free - of any diurnal cycle) over the Mediterranean Sea (MED) and the Black Sea (BS), respectively at 0.05° grid and covering the period from January 1st 1982 to present. The time series are regularly extended in time (possibly up to one month before real time) based on the availability of the upstream data.

Both MED and BS SST products are built from a consistent reprocessing of merged single-sensor Level-3C (L3C) climate data record (v.3.0), provided by the European Space Agency (ESA) Climate Change Initiative (CCI) and the Copernicus Climate Change Service (C3S) (Embury et al., 2024), but also include as input an adjusted version of the Advanced Very-High Resolution Radiometer (AVHRR) Pathfinder dataset version 5.3 (Saha et al., 2018) to increase the input observation coverage. Technical details on these products are available in the Product User Manual (PUM[2]) and in Pisano et al. (2016).

These products replace the previous reprocessed versions, which were based on an earlier version (v.2.1) of the European Space Agency (ESA) Climate Change Initiative (CCI) and Copernicus Climate Change Service (C3S) SST data (see also section VI).

The content of this document is an assessment based on the best set of in situ observations available for evaluation at the time of the product release. The validation methodology was

[1] 'Daily' indicates that data are delivered to the Copernicus Marine Data Store (MDS) each day, albeit with a delay of one month relative to real time, contingent on the availability of upstream data.

[2] https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-SST-PUM-010-021-022-041-042.pdf

defined and agreed within CMEMS, inheriting the long experience of the Marine Environment and security for the European Area (MERSEA) and MyOcean series of projects (Hernandez et al., 2018).

The baseline for the validation of these products is to use drifting buoy measurements only when they are available, as recommended by the Group for High Resolution SST (GHRSST) on satellite SST validation (Donlon et al., 2004). When the number of drifters' observations is low, as for the Black Sea, Argo floats, mooring buoys, and conductivity temperature depth probes (CTDs) measurements are used too. The assessment is based on the compilation of a matchup database between satellite SST and in situ data. The estimated accuracy numbers presented here such as the mean difference (mean bias) and the root mean square difference (RMSD) are computed on the matchups over a period of 17 years from 2005 to 2021 (see also Pisano et al., 2016). Differences are always computed as satellite SST minus in situ data.

I.2 Summary of the results

The accuracy of the MED and BS REP products (Table 1) has been assessed by comparison to independent and co-located in situ data (matchups). Validation covers the period 2005-2021 and quantifies the mean bias and RMSD from temperature differences.

For the Mediterranean Sea, only surface drifting buoys have been used, since their temporal and spatial coverage was rather homogeneous over the whole period. For the Black Sea, since the number of drifters was quite low, Argo floats, mooring buoys, and CTD data have been integrated (along with drifters' data). All in situ data are provided by the CMEMS INSITU (INS) Thematic Assembly Centre (TAC)[3].

Mean bias and RMSD for the MED-REP-L4 SST and MED-REP-L3S products have been quantified as 0.038 ± 0.001 K and 0.385 ± 0.001 K, and 0.012 ± 0.001 K and 0.387 ± 0.001 K respectively (Table 2). The Mean bias and RMSD for the BS-REP-L4 SST and BS-REP-L3S products have been quantified as 0.20 ± 0.01 K and 0.50 ± 0.01 K, and 0.18 ± 0.01 K and 0.50 ± 0.01 K respectively (Table 2).

The error of Black Sea L3S/L4 SST data, though greater than that of Mediterranean ones, is around half degree Kelvin. The difference with respect to the Mediterranean Sea can depend, at least in part, on the fact that the statistics in the Black Sea are based on different types of in situ instruments and on a much lower number of in situ measurements compared to the Mediterranean Sea.

[3] https://data.marine.copernicus.eu/product/INSITU_GLO_PHY_TS_DISCRETE_MY_013_001/description; https://data.marine.copernicus.eu/product/INSITU_GLO_PHYBGCWAV_DISCRETE_MYNRT_013_030/description

I.3 Estimated Accuracy Numbers

Table 2 shows summary of statistics of the differences between MED and BS REP data and co-located in situ measurements.

Table 2. *Summary statistics of the comparison (i.e., differences) between MED and BS REP SST with in situ data over the matchup points. Mean bias (K), Root Mean Square Difference (RMSD; K) and number of matchups were computed over the period 2005-2021. Each statistical parameter is associated with a 95 % confidence interval computed through a bootstrap procedure (Efron & Tibshirani, 1993).*

Dataset	In Situ data	Mean bias (K)	RMSD (K)	Matchups
MED-REP-L4	Drifting buoys	0.038 ± 0.001	0.385 ± 0.001	417780
MED-REP-L3S	Drifting buoys	0.012 ± 0.001	0.387 ± 0.001	417780
BS-REP-L4	Drifting buoys, Argo floats, mooring buoys, CTD	0.20 ± 0.01	0.50 ± 0.01	7295
BS-REP-L3S	Drifting buoys, Argo floats, mooring buoys, CTD	0.18 ± 0.01	0.50 ± 0.01	7295

II PRODUCTION SYSTEM DESCRIPTION

Production centre name: CNR-ISMAR, IT (Italy)

Production system name: Mediterranean Sea and Black Sea SST Reprocessing. The list of the four reprocessed products is shown in Table 1.

II.1 SST Processing chain and algorithms

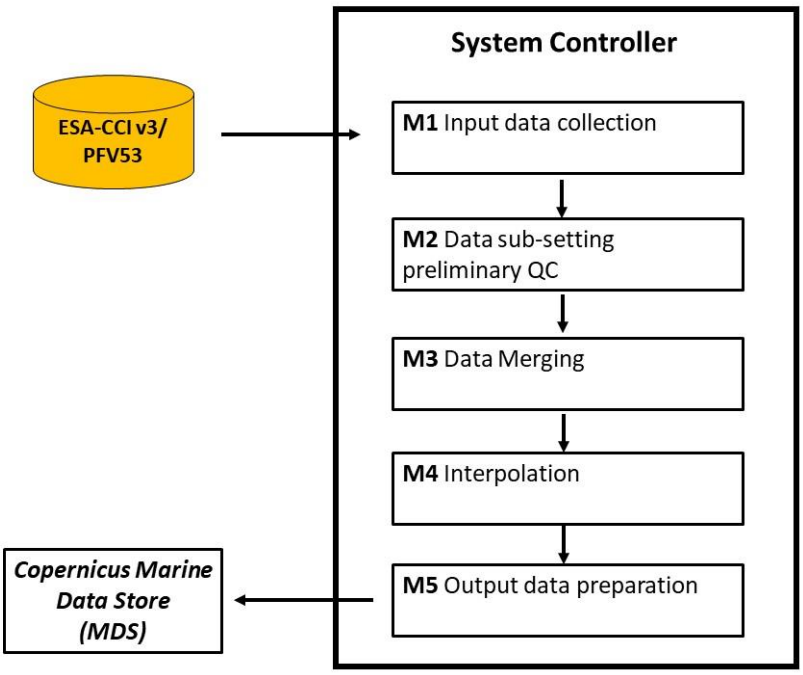

Figure 1. *Schematic diagram of the processing chain used at CNR-ISMAR for producing MED and BS REP L3S/L4 products. These products are built by using the same upstream data source, i.e. ESA CCI SST v.3.0 and Pathfinder v5.3 (PFV53[4]), and disseminated through the Copernicus Marine Data Store (MDS). The whole chain is structured into various modules (referred to as M1, M2, …, M5), which are outlined here as 'logical' steps applicable to both MED and BS SST processing.*

The regional reprocessed products provide daily (night time), merged multi-sensor (L3S), and optimally interpolated (L4) satellite-based estimates of the foundation SST over the Mediterranean Sea (MED) and Black Sea (BS), respectively at 0.05° grid and covering the period from January 1st 1982 to present. "Present" refers to approximately one month behind real time, though this timeliness depends on the effective availability of upstream data (see section II.1.1).

Both MED and BS SST products are built from a consistent reprocessing of the L3C climate data record (v.3.0), provided by the ESA Climate Change Initiative (CCI) and the Copernicus Climate

[4] https://www.ncei.noaa.gov/access/metadata/landing-page/bin/iso?id=gov.noaa.nodc:AVHRR_Pathfinder-NCEI-L3C-v5.3

Change Service (C3S) (Embury et al., 2024) but also include, as input, an adjusted version of the AVHRR Pathfinder dataset version 5.3 (Saha et al., 2018) to increase the input observation coverage.

The interpolation of the L3C upstream data to L4 has been carried out by adapting the CNR-ISMAR-GOS near-real-time (NRT) SST processing chain (Buongiorno Nardelli et al., 2013) following the approach described in Pisano et al. (2016). The operational chain is organized in various modules/packages (Figure 1), which are managed through a specific System Controller. For reprocessing purposes, the System Controller, which generally governs the sequence of operations, has been simplified with respect to the NRT chain, considering for example that the upstream data are already available as collated (merged) daily images (L3C) and setting aside the need for processing from Level 2 (L2) to Level 3.

II.1.1 Input upstream data

The CCI SST Climate Data Record (CDR) v.3.0 is the latest release from the ESA CCI SST project (Embury et al. 2024). Covering the period from 1980 to 2021, this data record has been extended as an Interim CDR (ICDR) since 2022, providing ongoing updates with a delay of approximately one month. Both SST CDRs and ICDRs are produced using software and algorithms developed through the ESA SST CCI project. CDR v.3 is available at the Centre for Environmental Data Analysis (CEDA[5]), while ICDR v.3 at the UK Group Workspace[6].

Both CDR and ICDR provide global daily SST fields at different processing levels, including L2P, L3U and L3C (Table 3). These data are produced using different satellite infrared and microwave sensors (Figure 2), i.e. the AVHRR (Advanced Very High Resolution Radiometer), (A)ATSR (Advanced Along-Track Scanning Radiometer), SLSTR (Sea and Land Surface Temperature Radiometer) and AMSR (Advanced Microwave Scanning Radiometer) series of sensors (Embury et al., 2024). The resulting data (Table 3) provides a stable, low-bias climate data record.

Following the public release of CCI SST v.3.0 in April 2024, a full reprocessing of the MED and BS data has been carried out. For this purpose, L3C maps derived from infrared sensors only, providing daily (nighttime) SST at 20 cm depth on a 0.05° x 0.05° latitude-longitude grid, have been used as input data. No microwave data were used. To note that ATSR data have not changed since the previous version, so v2.1 data have been used for the whole period available.

The Pathfinder v.5.3 (PFV53) dataset provides global daily (nighttime and daytime) L3C skin SST data at 4 km resolution, built from AVHRR observations aboard NOAA polar-orbiting satellites, and covering the period 1981 to present (Saha et al., 2018).

PFV53 data have been used to improve the number of SST observations during the period from late 1981 to 2014 (Table 3). Indeed, though the NOAA AVHRR sensors used in PFV53 are the same used in ESA CCI v.3.0, the latter occasionally showed an extremely low number of valid observations compared to PFV53. This discrepancy is due to the Quality Control (QC) flags used in deriving the ESA CCI L3C SST product. Data voids sometimes led to extremely degraded

[5] https://data.ceda.ac.uk/neodc/eocis/data/global_and_regional/sea_surface_temperature/CDR_v3
[6] https://gws-access.jasmin.ac.uk/public/cds_c3s_sst/data/ICDR_v3/

analyses, especially on a few months over the early period of the time series. Here, PFV53 data have thus been included after being bias-adjusted to the ESA CCI L3C data which are used as reference data.

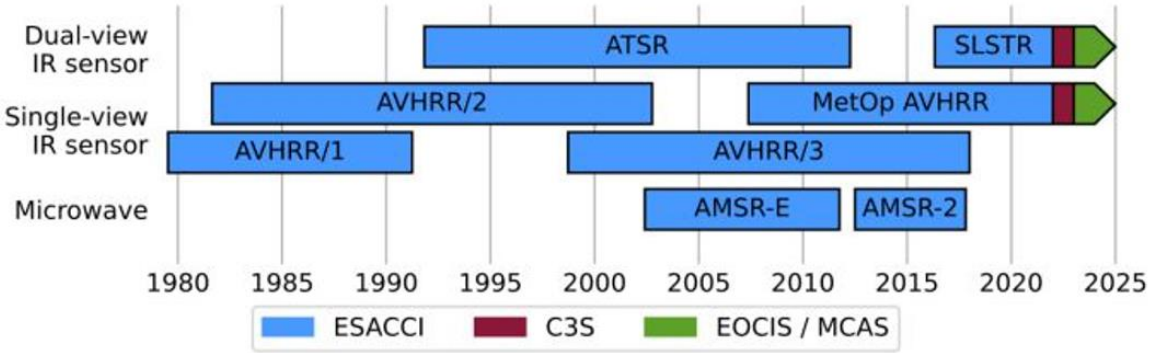

Figure 2. *Overview of products included in the Climate Change Initiative (CCI) SST CDR v3.0. Data from 2022 onwards are Interim-CDR (ICDR) funded by the Copernicus Climate Change Service (C3S) (2022), and the UK Earth Observation Climate Information Service (EOCIS) and UK Marine and Climate Advisory Service (MCAS) (2023 onwards). Both CDR and ICDR provide global daily SST fields at different processing levels, including L2P, L3U and L3C. These data are produced using different satellite infrared and microwave sensors, i.e. the AVHRR (Advanced Very High Resolution Radiometer), (A)ATSR (Advanced Along-Track Scanning Radiometer), SLSTR (Sea and Land Surface Temperature Radiometer) and AMSR (Advanced Microwave Scanning Radiometer) series of sensors (Embury et al., 2024).*

Table 3. *Input upstream L3C data used in the interpolation of the Copernicus Marine Service Mediterranean Sea and Black Sea SST products.*

Type	Version	Date range	Region	Data levels
CDR	ESA CCI v.3.0	1982 – 2021	Global	L2P, L3U, L3C
ICDR	ESA CCI v.3.0	2022 - 1 month behind real time	Global	L3C
Pathfinder	v.5.3	1982-2014	Global	L3C

II.1.2 Algorithms

A synthesis of the modules (Figure 1) used for the creation of the Copernicus Mediterranean Sea and Black Sea SST REP data is provided in the following sub-sections (II.1.2.1-II.1.2.4). The user can also refer to Buongiorno Nardelli et al. (2013) and Pisano et al. (2016) for more detailed information.

II.1.2.1 Collection of inputs (M1)

This module (Figure 1) manages the external interfaces to get the upstream L3C data. It was configured to collect only nighttime L3C data, from 1982 to present for CCI SST data and from

1982 to 2014 for PFV53 data (Table 3). All L3C data are provided with additional information, including quality level flags and cloud flags (see GDS 2.0.5 Data Processing Specification[7]).

II.1.2.2 L3C data extraction, preliminary data selection and additional cloud screening (M2)

L3C SST data are first extracted based on the geographical coverage. To avoid diurnal warming contamination (Buongiorno Nardelli et al., 2005), the reprocessing system only selects the night time L3C data. L3C data preliminary selection is then performed, starting from the quality flags and confidence values associated with each pixel. Only high quality flag data (i.e. from a flag value of 3 to 5, see also section III) are retained. Selected valid data are then saved in an internal format over the original CCI SST grid. Two additional checks for cloud contamination are performed by flagging pixels that are colder than the previous day's L3C value by a fixed threshold of 1.5 °C, or colder than the corresponding climatological value by 5 °C (see Pisano et al., 2016).

II.1.2.3 Multi-sensor merging: L3C to L3S (M3)

The various L3C data, obtained from module M2, are combined into a single daily merged multi-sensor map, also called a super-collated (L3S) map. However, SST values from one sensor can differ significantly from those of another due to differences in sensor characteristics (such as number of bands, spectral resolution, scanning/viewing geometry) and the specific algorithms applied, which may vary in how they correct for atmospheric effects on brightness temperature measurements. Therefore, a bias adjustment procedure is applied to each L3C dataset relative to a reference sensor during the merging process.

The multi-sensor merging process relies on a pre-defined list of reference sensors (Table 4) that provide the highest quality SST measurements for given time periods. Sensors with SST values that do not require correction (e.g., AVHRR 7 from January 1982 to February 1985) are included in this reference list, and their data form the initial reference SST map. Additionally, a list of sensors requiring adjustment is maintained, which includes all remaining L3C data (see Embury et al., 2024). During merging, each new sensor's L3C data is sequentially ingested from this secondary list. For each pixel, only the first observation encountered is retained, ensuring that only the best sensor measurement at each location is used.

Before adding new data to the merged map, any large-scale biases between the incoming L3C data and the current reference map (i.e., the already merged pixels) are identified and corrected. This bias is calculated over the overlapping areas. The merged SST map is updated with each new valid observation and subsequently serves as the reference for the next step.

[7] https://www.ghrsst.org/resources/ghrsst-project-documents/

Table 4. *List of reference sensors used in the multi-sensor merging procedure (module M2) to create the Copernicus MED and BS L3S REP data.*

Reference satellite sensors (from ESA CCI SST v.3.0 and C3S)	Start time (dd/mm/year)	End time (dd/mm/year)
AVHRR 7	01/01/1982	18/02/1985
AVHRR 9	19/02/1985	12/10/1988
AVHRR 11	13/10/1988	31/12/1991
ATSR1	01/01/1992	31/07/1995
ATSR2	01/08/1995	30/11/1995
AVHRR 14	01/12/1995	30/06/1996
ATSR2	01/07/1996	31/07/2002
AATSR	01/08/2002	31/03/2012
AVHRR METOP-A	01/04/2012	30/09/2021
SLSTR A	01/01/2017	present

II.1.2.4 Optimal interpolation algorithm: L3S to L4 (M4)

The same optimal interpolation (OI) scheme has been adopted for both the Mediterranean Sea and Black Sea SST reprocessing chains, except for the choice of the covariance models.

L4 analyses are obtained as a linear combination of the observations (i.e. the SST anomalies with respect to a background field), weighted directly with their correlation to the interpolation points and inversely with their cross-correlation and error. OI theory assumes that the statistical characteristics of the parameter variability (i.e., background error covariance and observation error covariance) are known. While several operational systems are based on purely spatial OI algorithms, using daily increments as input data (i.e. using previous day analysis as first guess, and eventually relaxing to a climatology in the absence of new observations, e.g., Donlon et al., 2012, and references therein), CNR scheme directly applies a space-time OI.

For both the MED and BS L4 production, the chosen OI scheme uses a daily climatology as first guess. The MED and BS climatology fields are derived from the v.3.0 SST CCI Climatology Data Record (CDR)[8] that provides daily global climatological mean SST derived from the CCI SST analysis data for the period 1991 to 2020 (30 years). The choice of this climatological period follows the recommendations of the World Meteorological Organization (WMO[9]).

Regarding the definition of the covariance function used in the OI scheme, the MED processing employs a spatially varying covariance function as described in Buongiorno Nardelli et al. (2015). In contrast, the BS processing uses a fixed spatial covariance function, based on Buongiorno Nardelli et al. (2010). The decision to use a fixed covariance for the Black Sea is due to the extremely high cloud coverage in the region, which makes it particularly challenging to derive complex covariance models directly from the available observations.

[8] https://data.ceda.ac.uk/neodc/eocis/data/global_and_regional/sea_surface_temperature/CDR_v3/Climatology/L4/v3.0.1

[9] https://wmo.int/media/news/updated-30-year-reference-period-reflects-changing-climate#:~:text=WMO%27s%20recent%20Services%20Commission%20meeting,switched%20to%20the%20new%20baseline.

The data sub-sampling strategy, inversion technique, and numerical implementation of the OI scheme are the same as those used in the NRT processing chain. In practice, the input data are selected only within a limited sub-domain (within a space-time influential radius) within a temporal window of ten days and a spatial influential radius, ranging between 300 and 900 km. To avoid data propagation across land, the REP scheme adopts a specific input data selection check. In particular, given the interpolation point and the input point selected within the sub-domain, the latter is considered for the interpolation only if there is no land pixel between the straight line connecting the two points.

The interpolation error (analysis error) is obtained from the formal definition of the error variance derived from OI theory (e.g. Bretherton et al. 1976).

III VALIDATION FRAMEWORK

The baseline for the validation of the MED and BS REP SST products is to use surface drifting buoy measurements only, when they are available, as recommended by the Group for High Resolution SST (GHRSST) on satellite SST validation (Donlon et al., 2004). When the number of drifters' observations is too low to allow significant metrics to be computed (as for the Black Sea), Argo floats, mooring buoys, and conductivity temperature depth probes (CTDs) measurements are used too.

The validation is based on the compilation of a matchup database between satellite SST products and in situ measurements. All the in situ data used are those made available by the Copernicus Marine Service INSITU (INS) TAC (see also section I.2). The INS TAC collects and distributes a variety of physical and biogeochemical seawater measurements provided with a homogeneous file format. Each in situ measurement undergoes an automated quality check before its distribution. Data quality is indicated by control flags ranging from 0 to 9, where a value of 0 means no quality control (QC) was performed, 1 represents the highest quality, and values from 2 to 4 indicate the progression from good to bad quality (for more details, see the QUID[10]). . For the purposes at hand, temperature measurements from surface drifting buoys, ARGO floats, mooring buoys and conductivity temperature depth probes (CTDs) were collected from the INS-TAC historical archive (these are indeed the in situ instruments that provide water temperature measurements). It should be noted that surface drifting buoys were only used for the Mediterranean Sea, while all other available in situ data, including drifters, were used for the Black Sea.

More in detail, the matchup database was built based on the following approach: first, a pre-selection of high-quality in situ data was performed, retaining only temperatures with quality flag equal to 1 (good) or 2 (probably good). Then, the co-location was carried out on a daily basis by collecting the closest (nearest neighbour) SST (for both L3S and L4) grid points to the in situ measurement within a temporal window spanning from 21:00 local time of the previous day) to 06:00 local time of the validating day. The overall validation covers the period from 1st January 2005 to 31st December 2021.

Before computing the statistics of the differences between satellite-based SST and in situ estimates, specific QC procedures (i.e. additional to those provided by the INS-TAC) are applied to the in situ data. In fact, in situ sensors may be affected by significant systematic errors or may experience malfunctioning as a consequence of a variety of factors (lost drogue, broken sensor, etc.). In this context, only the in situ data that passed all QC tests were included in the matchup database. These additional tests include a comparison between in situ data and the daily climatological SST field that is used to compute the SST L4 anomalies (rejecting all in situ values exceeding a difference of 5 K), and a comparison with the previous day L4 product; thereby rejecting all in situ values exceeding a difference of 1.5 K, which is the same criterion used for satellite L2P data rejection in the CNR-ISMAR operational NRT L4 processing chain (see also Pisano et al., 2016).

[10] https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-INS-QUID-013-001.pdf

IV VALIDATION RESULTS

IV.1 Mediterranean Sea

Tables 5 and 6 summarize the main statistical parameters (i.e., mean bias and RMSD with respect to in situ data obtained for the MED-REP-L4 and MED-REP-L3S products respectively). Only drifters' data are used for the assessment of this product and the validation covers the period 2005-2021.

Table 5. *Summary statistics of the differences between MED-REP-L4 SST and drifters' data over the matchup points. Mean bias (K), Root Mean Square Difference (RMSD; K) and number of matchups were computed over the period 2005-2021. Each statistical parameter is associated with a 95 % confidence interval computed through a bootstrap procedure (Efron & Tibshirani, 1993).*

Dataset Name: cmems_SST_MED_SST_L4_REP_OBSERVATIONS_010_021			
Start Date	**End Date**	**Parameter**	**Value**
2005-01-27	2021-12-31	Mean bias	0.038 ± 0.001K
2005-01-27	2021-12-31	RMSD	0.385 ± 0.001 K
2005-01-27	2021-12-31	Matchup number	417780

Table 6. *Summary statistics of the differences between MED-REP-L3S SST and drifters' data over the matchup points. Mean bias (K), Root Mean Square Difference (RMSD; K) and number of matchups were computed over the period 2005-2021. Each statistical parameter is associated with a 95 % confidence interval computed through a bootstrap procedure (Efron & Tibshirani, 1993).*

Dataset Name: cmems-obs-sst-med-phy-my-l3s-P1D-m			
Start Date	**End Date**	**Parameter**	**Value**
2005-01-27	2021-12-31	Mean bias	0.012 ± 0.001 K
2005-01-27	2021-12-31	RMSD	0.387 ± 0.001 K
2005-01-27	2021-12-31	Matchup number	417780

The spatial distribution of the matchup points (i.e. the co-located L3S, L4 and drifters' data in Figure 3) shows a homogeneous coverage over the entire MED-REP-L4/L3S geographical domain, although an uneven in situ data coverage characterizes the North Adriatic and North Aegean Sea.

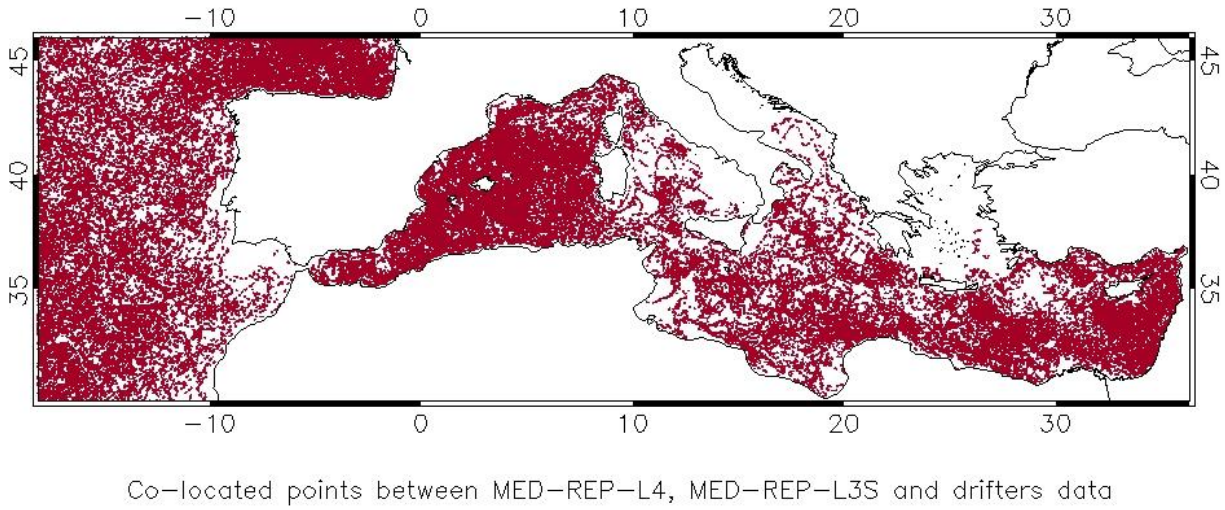

***Figure 3**. Spatial distribution of the matchup points over the MED-REP-L4/L3S domain from 2005 to 2021. Each point represents the position where MED-REP-L4, MED-REP-L3S and drifter data are all available.*

Looking at the spatial distribution of the mean bias, both MED-REP-L3S and MED-REP-L4 show similar patterns (Figure 4): the bias is nearly zero, remaining below 0.05 °C across most of the Mediterranean Sea, while small negative values (MED-REP-L4/L3S SSTs colder than drifters' temperatures) are present in the adjacent Atlantic area.

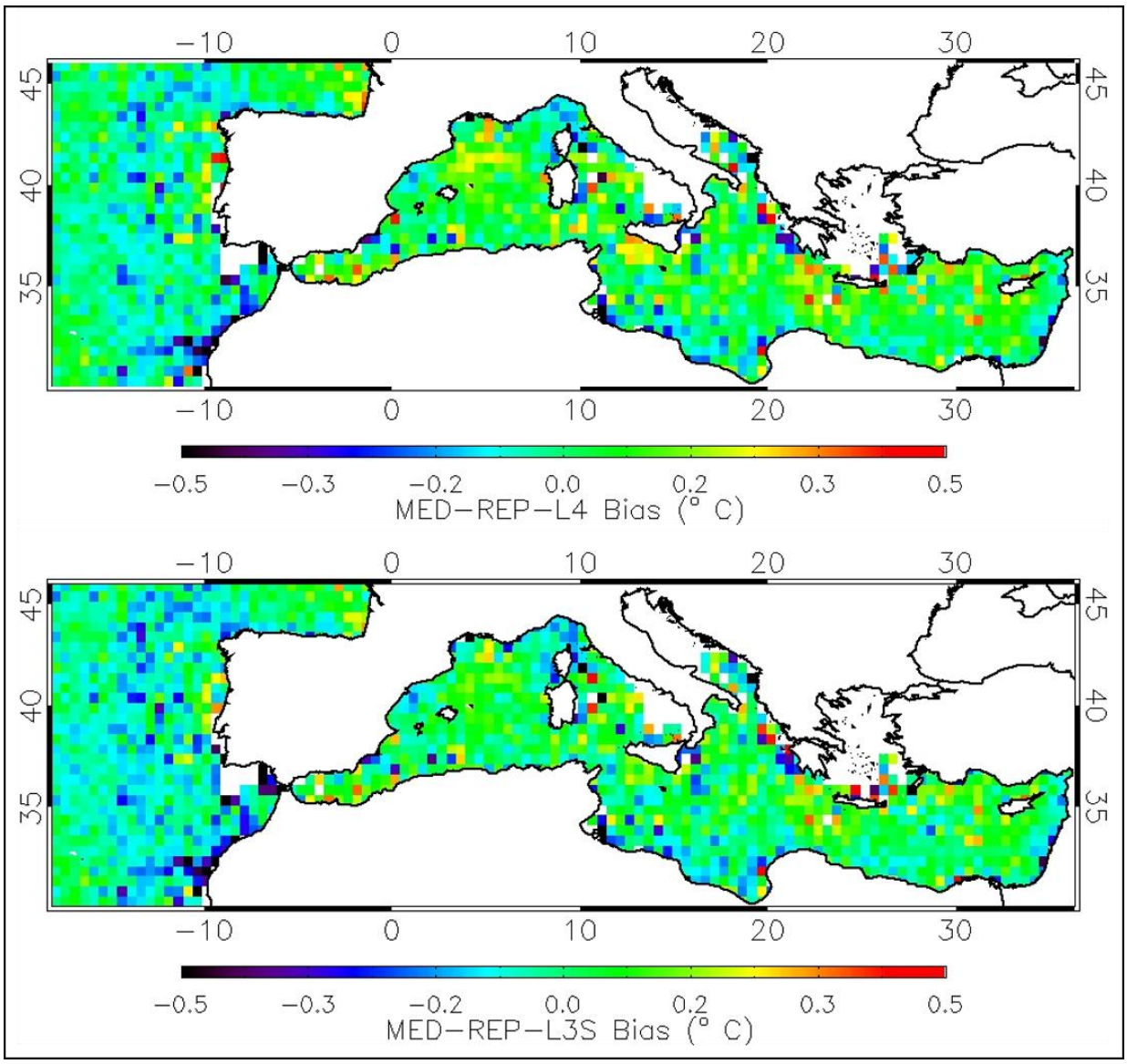

***Figure 4**. Spatial map of mean bias for the MED-REP-L3S (upper panel) and MED-REP-L4 (lower panel) products computed over the matchup database from 2005 to 2021. Mediterranean domain.*

Similar considerations are for the spatial distribution of the RMSD (Figure 5) where both MED-REP-L3S and MED-REP-L4 show similar patterns that shows a RMSD variability between 0.2 and 0.5 K.

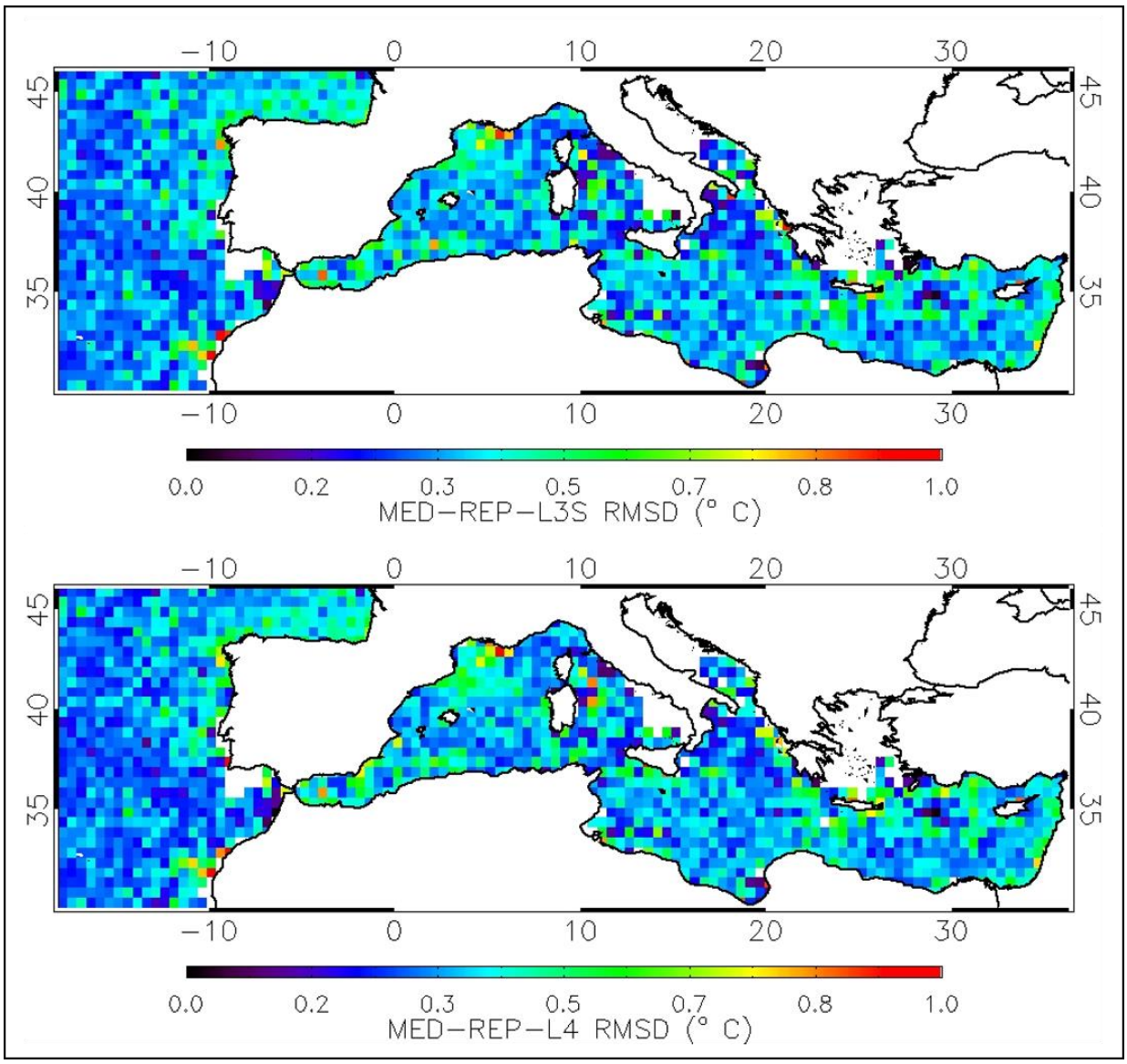

***Figure 5**. Spatial map of Root Mean Square Difference (RMSD) for the MED-REP-L3S (upper panel) and MED-REP-L4 (lower panel) products computed over the matchup database from 2005 to 2021. Mediterranean domain.*

The time series of yearly averaged mean bias and RMSD of both MED-REP-L4 and MED-REP-L3S products (Figure 6 and 7) show very good stability over the years, that is neither the bias or RMSD show any significant trend. Indeed, no slopes are seen in the yearly averaged time series shown in Figures 6 and 7, with values of 0.005 ± 0.0005 K/year for the bias and -0.001 ± 0.0005 K/year for the RMSD (uncertainty of the slopes is given at 95% confidence level). Thus, we can safely exclude any systematic SST deterioration or spurious trend in the data. The temporal stability of MED REP L3S and L4 products ensures the appropriateness of their usage for climate studies.

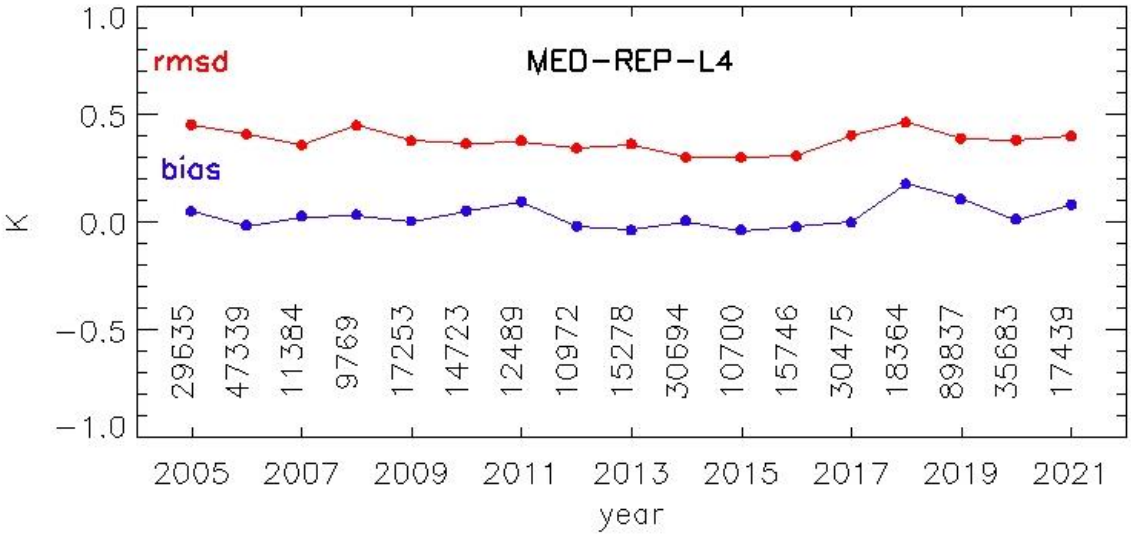

Figure 6. *Yearly averaged RMSD (red line) and mean bias (blue line) of the differences (K) between MED-REP-L4 SSTs and drifters' data over the period 2005-2021.*

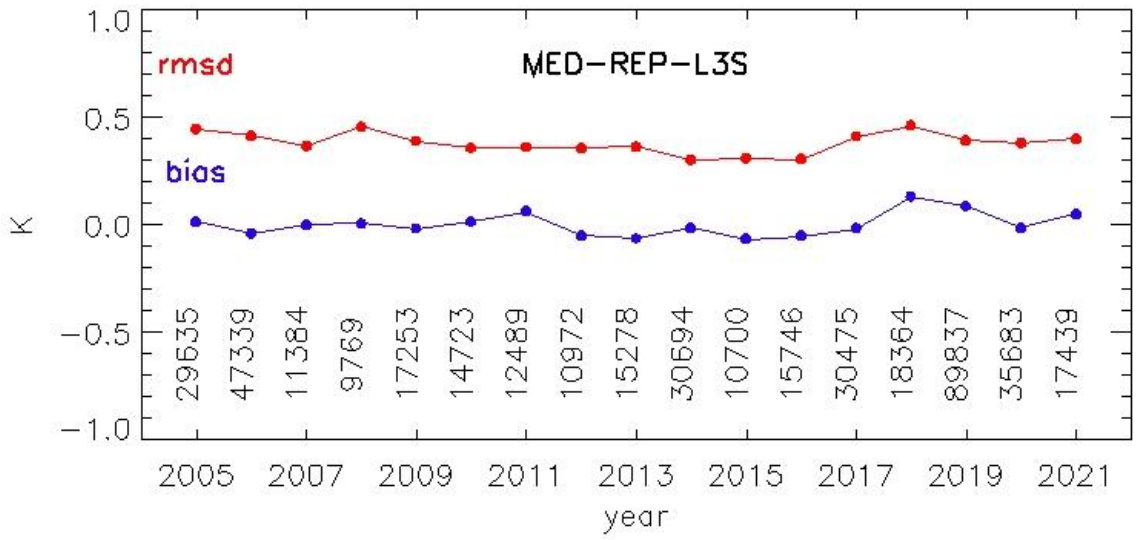

Figure 7. *Yearly averaged RMSD (red line) and mean bias (blue line) of the differences (K) between MED-REP-L3S SSTs and drifters' data over the period 2005-2021.*

The density scatter plots of MED-REP-L4 and MED-REP-L3S products (Figure 8) give another picture of the overall statistical analysis. Both MED-REP-L4 and MED-REP-L3S scatter plots show a very good agreement with respect to in situ drifter observations (Pearson correlation coefficient, r = 0.99).

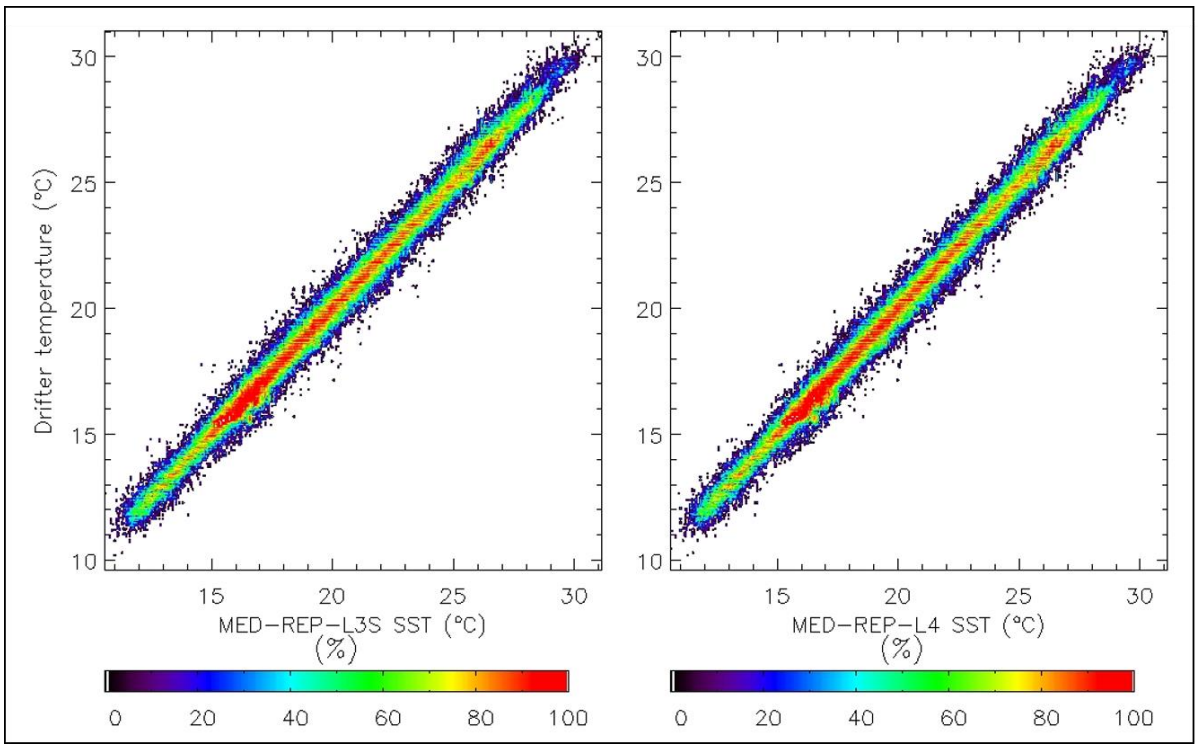

***Figure 8**. Density scatter plots of drifters' data vs. MED-REP-L3S (left panel) and MED-REP-L4 (right panel) SSTs over the period 2005-2021. The color bar indicates the percentage relative to the total number of matchups (Tables 5 and 6). The estimated Pearson correlation is 0.99.*

IV.2 Black Sea

Tables 7 and 8 summarize the main statistical parameters (i.e. mean bias and RMSD with respect to in situ data obtained for the BS-REP-L4 and BS-REP-L3S products respectively). Drifters, Argo float, mooring buoys and CTD data are used for the assessment of this product and the validation covers the period 2005-2021.

***Table 7.** Summary statistics of the differences between BS-REP-L4 SST and in situ data over the matchup points. Mean bias (K), Root Mean Square Difference (RMSD; K) and number of matchups were computed over the period 2005-2021. Each statistical parameter is associated with a 95 % confidence interval computed through a bootstrap procedure (Efron & Tibshirani, 1993).*

Dataset Name: cmems_SST_BS_SST_L4_REP_OBSERVATIONS_010_022

Start Date	End Date	Parameter	Value
2005-01-31	2021-12-31	Mean bias	0.20 ± 0.01K
2005-01-31	2021-12-31	RMSD	0.50 ± 0.01K
2005-01-31	2021-12-31	Matchup number	7295

Table 8. *Summary statistics of the differences between BS-REP-L3S SST and in situ data over the matchup points. Mean bias (K), Root Mean Square Difference (RMSD; K) and number of matchups were computed over the period 2005-2021. Each statistical parameter is associated with a 95 % confidence interval computed through a bootstrap procedure (Efron & Tibshirani, 1993).*

Dataset Name: cmems-obs-sst-bs-phy-my-l3s-P1D-m

Start Date	End Date	Parameter	Value
2005-01-31	2021-12-31	Mean bias	0.18 ± 0.01K
2005-01-31	2021-12-31	RMSD	0.50 ± 0.01K
2005-01-31	2021-12-31	Matchup number	7295

The spatial distribution of the matchup points is shown in Figure 9. Although the spatial coverage is rather uneven in the Black Sea, the number of observations is acceptable for statistical analysis.

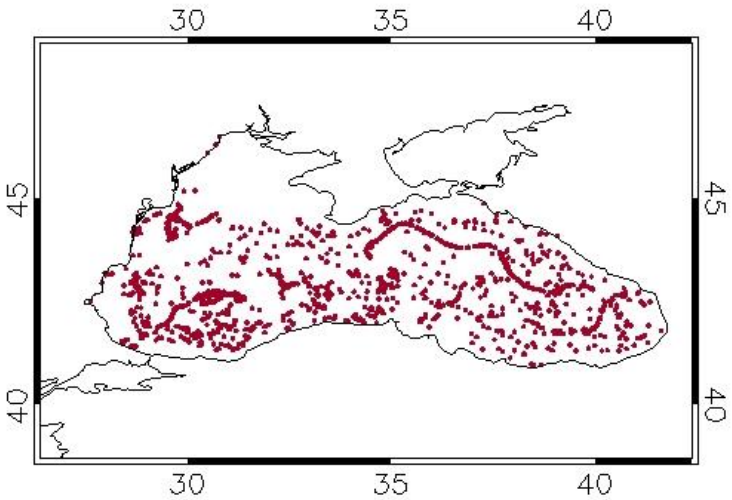

Figure 9*. Spatial distribution of the matchup points over the BS-REP-L4/L3S domain from 2005 to 2021. Each point represents the position where BS-REP-L4, BS-REP-L3S and drifter data are all available. Black Sea region.*

Looking at the spatial distribution of the mean bias, both BS-REP-L3S and BS-REP-L4 show similar patterns (Figure 10); the bias is rather uniform over the entire Black Sea domain ranging around 0.2 K.

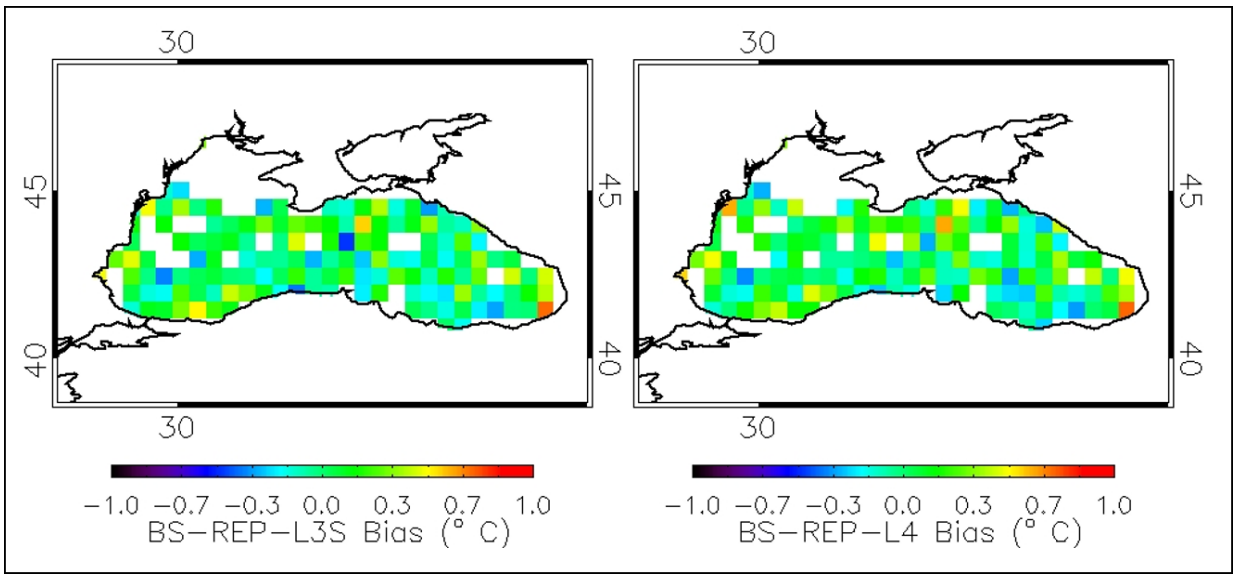

***Figure 10**. Spatial map of mean bias for the BS-REP-L3S (left panel) and BS-REP-L4 (right panel) products computed over the matchup database from 2005 to 2021. Black Sea region.*

Similar considerations are for the spatial distribution of the RMSD (Figure 11) where both BS-REP-L3S and BS-REP-L4 show similar patterns that evidence a RMSD variability between 0.3 and 0.7 K.

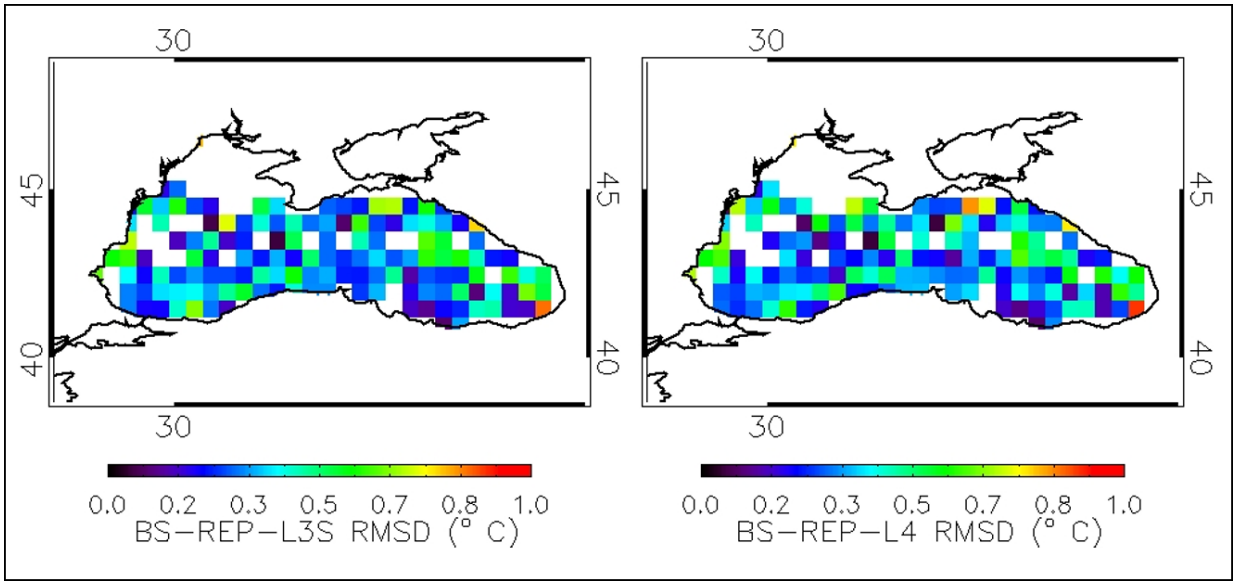

***Figure 11**. Spatial map of RMSD for the BS-REP-L3S (left panel) and BS-REP-L4 (right panel) products computed over the matchup database from 2005 to 2021. Black Sea region.*

Overall, both the bias and RMSD are greater than those found for the Mediterranean Sea product. This can depend, at least in part, on the fact that the statistics in the Black Sea are based on different types of in situ instruments and on a much lower number of in situ

measurements compared to the Mediterranean Sea. We recall that drifters provide temperature measurements at about 20 cm below the sea surface while Argo floats and moored buoys at lower depths.

The time series of yearly averaged mean bias and RMSD of BS-REP-L4 (Figure 12) shows a slight positive tendency for both the bias and RMSD, with the same linear slope of 0.01 ± 0.003K/year (at 95% confidence level). Analogous results are for the BS-REP-L3S bias and RMSD time series (Figure 13), with the same linear slope of 0.01 ± 0.002 K/year. However, some years are not covered by data while others (2014) have very few observations.

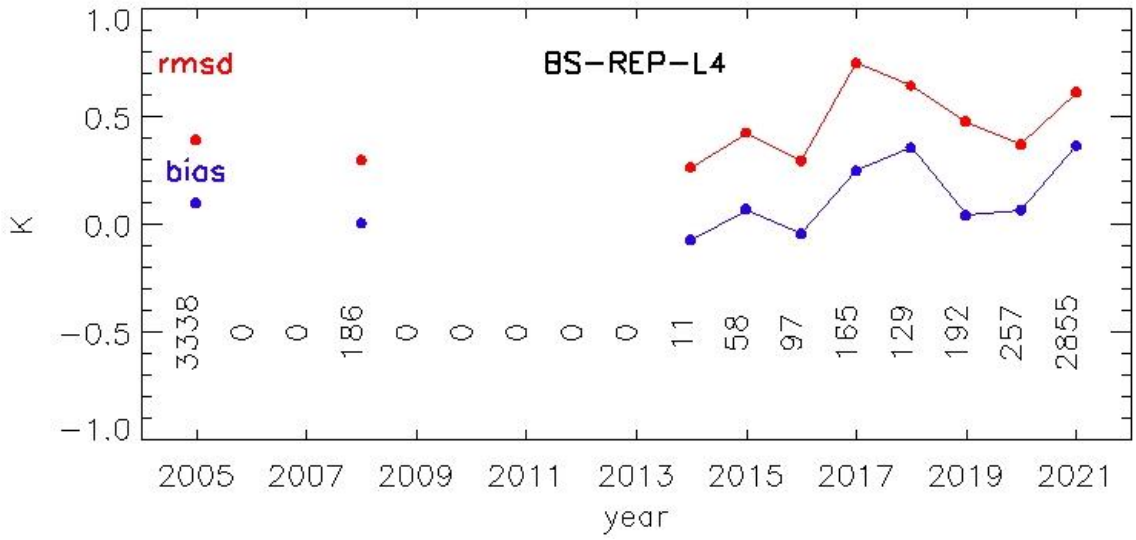

Figure 12. *Yearly averaged Root Mean Square Difference (RMSD) (red line) and mean bias (blue line) of the differences (K) between BS-REP-L4 SSTs and in situ data over the period 2005-2021.*

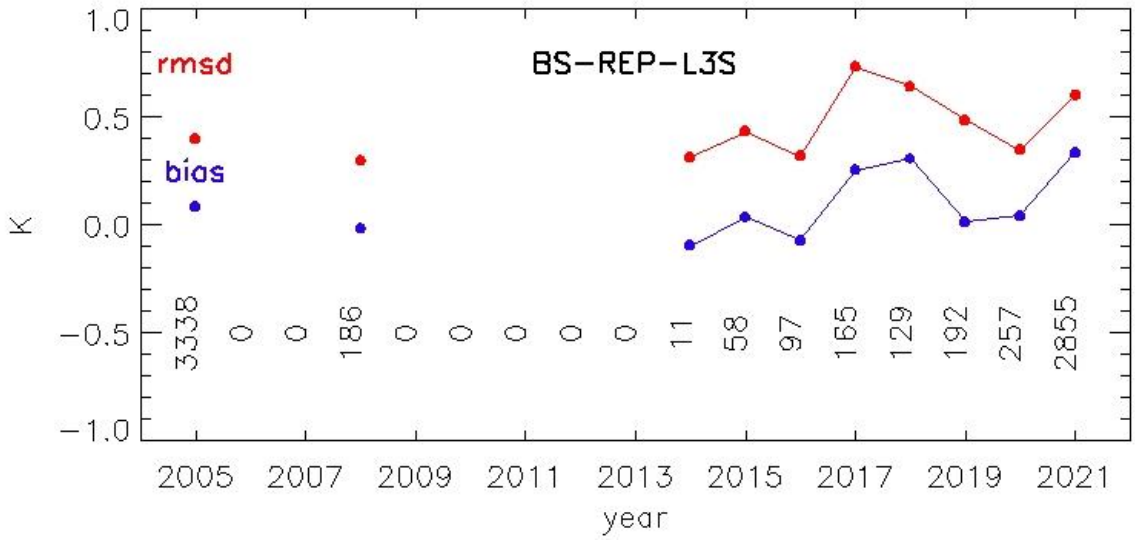

Figure 13. *Yearly averaged Root Mean Square Difference (RMSD) (red line) and mean bias (blue line) of the differences (K) between BS-REP-L3S SSTs and in situ data over the period 2005-2021.*

The density scatter plots of both BS-REP-L3S and BS-REP-L4 (Figure 14) give another picture of the overall statistical analysis. Both L3S and L4 SST data show a good agreement with respect to in situ drifter observations (Pearson correlation coefficient, r = 0.99).

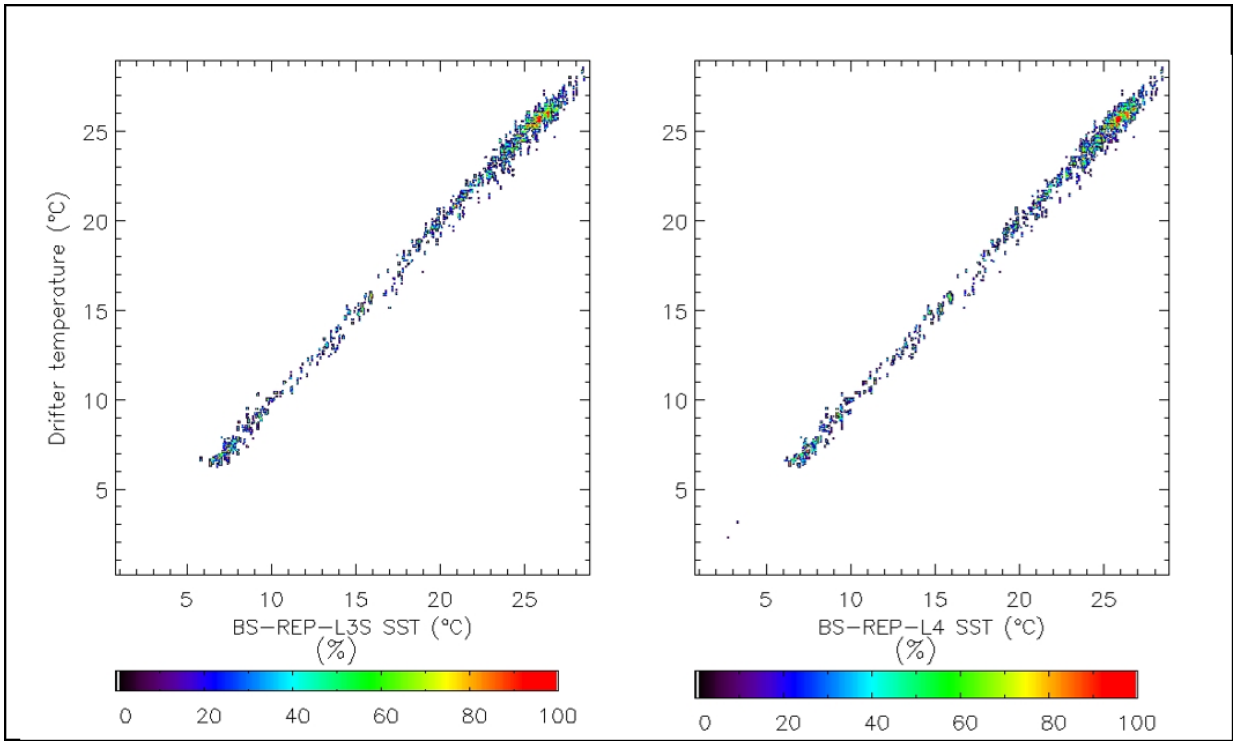

Figure 14. *Density scatter plots of drifters' data vs. BS-REP-L3S (left panel) and BS-REP-L4 (right panel) SSTs over the period 2005-2021.* The color bar indicates the percentage relative to the total number of matchups (Tables 7 and 8). *Pearson correlation coefficient is 0.99.*

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

This section summarizes the main changes occurred to the Mediterranean Sea and Black Sea Reprocessed products from 2022 onward.

Date	Change	Event description
11 April 2022	Upstream data change and switch to C3S data v.2.1.	Since 11 April 2022, the Mediterranean and Black Sea SST Reprocessed Products are produced using version 2.1 C3S data instead of v2.0 and MetOp-B AVHRR data instead of MetOp-A. Both changes are made at the same time, i.e. for SST REP data from October 2021 onwards. The integration of MetOp-B does not impact the quality of SST REP products, i.e. no changes in their quality.
16 June 2022	Release of MED-L3S-REP and BS-L3S-REP products.	This evolution introduces two new reprocessed products: the Mediterranean Sea and Black Sea L3S REP dataset, which complement the existing L4 REP dataset.
16 June 2023	Automatic temporal extension of MED and BS L3S and L4 products to reach one month before real time.	This evolution introduces an automatic temporal extension of the Mediterranean Sea and Black Sea reprocessed products in order to reach one month before real time.
31 May 2024	Upstream data change to ESA CCI data v.3.0.	Full reprocessing of the Mediterranean Sea and Black Sea multi-year products using ESA-CCI v.3.0.

VI QUALITY CHANGES SINCE PREVIOUS VERSION

To evaluate quality changes relative to the previous version of the reprocessed Mediterranean Sea and Black Sea products, we provide a summary of statistics for these discontinued products. Table 9 presents key statistics for this prior version of the SST datasets, derived from ESA CCI SST v2.1. Validation was performed using the same in situ data sample and time period (2005 to 2021) as for the current version, based on ESA CCI SST v.3.0. This allows for a straightforward comparison between previous (Table 9) and current (Table 2) versions.

Comparing Tables 2 and 9 reveals that the current Mediterranean Sea version shows a substantial reduction in RMSD relative to the previous version, while the bias remains largely unchanged. For the Black Sea, the current version displays a slight decrease in RMSD but a small increase in bias (see Table 2 for comparison). As discussed earlier, the difference between the Mediterranean and Black Seas may be partly due to the use of different types of in situ instruments in the Black Sea and the significantly lower number of in situ measurements available for validation compared to the Mediterranean Sea.

Table 9. *Summary statistics for the previous version of the reprocessed SST datasets for the Mediterranean Sea and Black Sea, based on CCI SST v.2.1. RMSD: Root Mean Square Difference. Each statistical parameter is associated with a 95 % confidence interval computed through a bootstrap procedure (Efron & Tibshirani, 1993).*

Dataset	In Situ data	Mean bias (K)	RMSD (K)	Matchups
MED-REP-L4	Drifting buoys	0.036 ± 0.001	0.456 ± 0.001	474025
MED-REP-L3S	Drifting buoys	0.008 ± 0.001	0.466 ± 0.001	474025
BS-REP-L4	Drifting buoys, Argo floats, mooring buoys, CTD	0.11 ± 0.01	0.50 ± 0.01	7149
BS-REP-L3S	Drifting buoys, Argo floats, mooring buoys, CTD	0.09 ± 0.01	0.51 ± 0.01	7149

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